

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250272519

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Eng; Adam D. et al.

RFID Tag Capable of Variable Information Based on Orientation

Abstract

Systems, apparatuses, and methods are described for a radio-frequency identification (RFID) tag that provides different data based on different orientations of the RFID tag. The RFID tag may include an orientation sensor such as, for example, a tilt switch that may detect an orientation of the RFID tag. When the RFID tag is determined to be in a first orientation (e.g., a vertical position), the RFID tag may provide a first data signal. When the RFID tag is determined to be in a second orientation (e.g., a horizontal position), the RFID tag may provide a second data signal. The first data signal and the second data signal may indicate different information to an RFID tag reader such as, for example, different identifications (IDs), different commands, and/or different requests.

Inventors: Eng; Adam D. (Golden, CO), Eng; David A. (Louisville, CO)

Applicant: Comcast Cable Communications, LLC (Philadelphia, PA)

Family ID: 1000007758988

Appl. No.: 18/590732

Filed: February 28, 2024

Publication Classification

Int. Cl.: G06K7/10 (20060101); G06K19/077 (20060101); H01Q1/22 (20060101)

U.S. Cl.:

CPC G06K7/10089 (20130101); G06K19/07773 (20130101); H01Q1/2225 (20130101);

Background/Summary

BACKGROUND

[0001] Conventional radio-frequency identification (RFID) tags are configured to always provide stored data when scanned by an RFID tag reader. The stored data is provided regardless of the position of the RFID tag, without possibilities of providing different types of data. Also, this may allow a bad actor to covertly obtain sensitive data stored on a conventional RFID tag. These and other shortcomings are addressed in the disclosure.

SUMMARY

[0002] The following summary presents a simplified summary of certain features. The summary is not an extensive overview and is not intended to identify key or critical elements.

[0003] Systems, apparatuses, and methods are described for a radio-frequency identification (RFID) tag that may provide different data based on different orientations of the RFID tag. The RFID tag may include an orientation sensor such as, for example, a tilt switch that may detect an orientation of the RFID tag. When the RFID tag is determined to be in a first orientation (e.g., a vertical position), the RFID tag may provide a first data signal based on first stored data. When the RFID tag is determined to be in a second orientation (e.g., a horizontal position), the RFID tag may provide a second data signal based on second stored data. The first data signal and the second data signal may indicate different information to an RFID tag reader such as, for example, different identifications (IDs), different commands, or different requests. Also, or alternatively, the first data signal may provide valid information while the second data signal may provide invalid information or null data. For example, the first data signal may indicate a valid authorization for a financial transaction while the second data signal may indicate that the financial transaction is not authorized. By providing the first data signal only when the RFID tag is in the first orientation, sensitive data stored on the RFID tag (e.g., data confirming authorization of a financial transaction) may be prevented from being provided by the RFID tag at certain times. As a result, a bad actor is less likely to obtain sensitive data stored on the RFID tag, thereby improving the security of the sensitive data. Further, the RFID tag may cause performance of different functions based on a user orienting the RFID tag differently to provide different data signals, expanding the capabilities and usefulness of the RFID tag.

[0004] These and other features and advantages are described in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Some features are shown by way of example, and not by limitation, in the accompanying drawings. In the drawings, like numerals reference similar elements.

[0006] FIG. 1 shows an example communication network.

[0007] FIG. 2 shows example components of a computing device.

[0008] FIG. 3 shows an example RFID tag.

[0009] FIG. 4A shows an example tilt switch in a first position.

[0010] FIG. 4B shows the example tilt switch of FIG. 4A in a second position.

[0011] FIG. 5 shows an example range of orientations of an example tilt switch.

[0012] FIG. 6 shows examples of different output signals provided by the RFID tag of FIG. 3 based on different corresponding orientations of the RFID tag.

[0013] FIG. 7 shows an example RFID tag that includes a temperature detection switch.

[0014] FIG. 8 shows an example RFID tag reader.

[0015] FIG. 9 shows example orientations of the RFID tag of FIG. 3 when embedded within a hand of a person.

[0016] FIG. 10 shows example user-specific orientations for activating the RFID tag of FIG. 3.

[0017] FIG. 11 is a flow chart showing steps of an example method of an RFID tag providing data

based on an orientation of the RFID tag.

[0018] FIG. 12 is a flow chart showing steps of an example method of an RFID tag reader receiving data from an RFID tag.

DETAILED DESCRIPTION

[0019] The accompanying drawings, which form a part hereof, show examples of the disclosure. It is to be understood that the examples shown in the drawings and/or discussed herein are non-exclusive and that there are other examples of how the disclosure may be practiced.

[0020] A radio-frequency identification (RFID) tag may provide different data based on different orientations of the RFID tag. For example, when the RFID tag is in a first orientation (e.g., a vertical orientation), the RFID tag may provide first data, and when the RFID tag is in a second orientation (e.g., a horizontal orientation), the RFID tag may provide second data. The first data and the second data may indicate different information such as, for example, different identification (ID) codes or different commands. For example, the first data may indicate authorization of a financial transaction while the second data may indicate that a financial transaction is not authorized. The RFID tag may include an orientation sensor such as, for example, a tilt switch or other orientation detection component to detect different orientations of the RFID tag. Based on a detected orientation of the RFID tag, the RFID tag may determine corresponding data to be provided by the RFID tag.

[0021] The data provided by the RFID tag may be provided to an RFID tag reader. The RFID tag reader may facilitate performance of an instruction indicated by the data (e.g., authorization of a financial transaction). The RFID tag reader may interrogate or scan the RFID tag by sending a signal that is received by the RFID tag. The RFID tag may provide data in response to the received signal based on the orientation of the RFID tag. The data may be provided via a modulated backscattered signal. The modulated backscattered signal may be based on the received signal from the RFID tag reader. The data provided to the RFID tag reader from the RFID tag, either individually or collectively as a sequence of provided data, may indicate various information such as, for example, an ID, a command, an authorization, a request, or other information.

[0022] By varying the data provided by the RFID tag based on the orientation of the RFID tag, bad actors may be prevented from surreptitiously scanning the RFID tag to obtain sensitive data from the RFID tag. That is, the RFID tag may only provide sensitive data when positioned in a certain specific orientation, thereby reducing the likelihood that a bad actor may covertly obtain the sensitive information. Further, the RFID tag may be used to enhance security by facilitating various multi-factor authentication (MFA) tasks such as, for example, authorizing a transaction, controlling a device (e.g., operating a vehicle), or providing verified access to a device (e.g., unlocking a vehicle).

[0023] FIG. 1 shows an example communication network 100 in which features described herein may be implemented. The communication network 100 may comprise one or more information distribution networks of any type, such as, without limitation, a telephone network, a wireless network (e.g., an LTE network, a 5G network, a Wi-Fi IEEE 802.11 network, a WiMAX network, a satellite network, and/or any other network for wireless communication), an optical fiber network, a coaxial cable network, and/or a hybrid fiber/coax distribution network. The communication network 100 may use a series of interconnected communication links 101 (e.g., coaxial cables, optical fibers, wireless links, etc.) to connect multiple premises 102 (e.g., businesses, homes, consumer dwellings, train stations, airports, etc.) to a local office 103 (e.g., a headend). The local office 103 may send downstream information signals and receive upstream information signals via the communication links 101. Each of the premises 102 may comprise devices, described below, to receive, send, and/or otherwise process those signals and information contained therein.

[0024] The communication links 101 may originate from the local office 103 and may comprise components not shown, such as splitters, filters, amplifiers, etc., to help convey signals clearly. The communication links 101 may be coupled to one or more wireless access points 127 configured to

communicate with one or more mobile devices **125** via one or more wireless networks. The mobile devices **125** may comprise smart phones, tablets or laptop computers with wireless transceivers, tablets or laptop computers in communication with other devices with wireless transceivers, and/or any other type of device configured to communicate via a wireless network.

[0025] The local office **103** may comprise an interface **104**. The interface **104** may comprise one or more computing devices configured to send information downstream to, and to receive information upstream from, devices communicating with the local office **103** via the communications links **101**. The interface **104** may be configured to manage communications among those devices, to manage communications between those devices and backend devices such as servers **105-107**, and/or to manage communications between those devices and one or more external networks **109**. The interface **104** may, for example, comprise one or more routers, one or more base stations, one or more optical line terminals (OLTs), one or more termination systems (e.g., a modular cable modem termination system (M-CMTS) or an integrated cable modem termination system (I-CMTS)), one or more digital subscriber line access modules (DSLAMs), and/or any other computing device(s). The local office **103** may comprise one or more network interfaces **108** that comprise circuitry needed to communicate via the external networks **109**. The external networks **109** may comprise networks of Internet devices, telephone networks, wireless networks, wired networks, fiber optic networks, and/or any other desired network. The local office **103** may also or alternatively communicate with the mobile devices **125** via the interface **108** and one or more of the external networks **109**, e.g., via one or more of the wireless access points **127**.

[0026] The push notification server **105** may be configured to generate push notifications to deliver information to devices in the premises **102** and/or to the mobile devices **125**. The content server **106** may be configured to provide content to devices in the premises **102** and/or to the mobile devices **125**. This content may comprise, for example, video, audio, text, web pages, images, files, etc. The content server **106** (and/or an authentication server) may comprise software to validate user identities and entitlements, to locate and retrieve requested content, and/or to initiate delivery (e.g., streaming) of the content. The application server **107** may be configured to offer any desired service. For example, an application server may be responsible for collecting, and generating a download of, information for electronic program guide listings. Another application server may be responsible for monitoring user viewing habits and collecting information from that monitoring for use in selecting advertisements. Yet another application server may be responsible for formatting and inserting advertisements in a video stream being transmitted to devices in the premises **102** and/or to the mobile devices **125**. The local office **103** may comprise additional servers, such as additional push, content, and/or application servers, and/or other types of servers. Also or alternatively, one or more servers may be part of the external network **109** and may be configured to communicate (e.g., via the local office **103**) with computing devices located in or otherwise associated with one or more premises **102** and/or with mobile devices **125**.

[0027] An example premises **102a** may comprise an interface **120**. The interface **120** may comprise circuitry used to communicate via the communication links **101**. The interface **120** may comprise a modem **110**, which may comprise transmitters and receivers used to communicate via the communication links **101** with the local office **103**. The modem **110** may comprise, for example, a coaxial cable modem (for coaxial cable lines of the communication links **101**), a fiber interface node (for fiber optic lines of the communication links **101**), twisted-pair telephone modem, a wireless transceiver, and/or any other desired modem device. One modem is shown in FIG. **1**, but a plurality of modems operating in parallel may be implemented within the interface **120**. The interface **120** may comprise a gateway **111**. The modem **110** may be connected to, or be a part of, the gateway **111**. The gateway **111** may be a computing device that communicates with the modem(s) **110** to allow one or more other devices in the premises **102a** to communicate with the local office **103** and/or with other devices beyond the local office **103** (e.g., via the local office **103** and the external network(s) **109**). The gateway **111** may comprise (and/or otherwise perform

operations of) a set-top box (STB), digital video recorder (DVR), a digital transport adapter (DTA), a computer server, a router, and/or any other desired computing device.

[0028] The gateway **111** may also comprise one or more local network interfaces to communicate, via one or more local networks, with devices in the premises **102a**. Such devices may comprise, e.g., display devices **112** (e.g., televisions), other devices **113** (e.g., a DVR or STB), personal computers **114**, laptop computers **115**, wireless devices **116** (e.g., wireless routers, wireless laptops, notebooks, tablets and netbooks, cordless phones (e.g., Digital Enhanced Cordless Telephone—DECT phones), mobile phones, mobile televisions, personal digital assistants (PDA)), landline phones **117** (e.g. Voice over Internet Protocol—VoIP phones), RFID tag reader **150**, and any other desired devices. Example types of local networks comprise Multimedia Over Coax Alliance (MoCA) networks, Ethernet networks, networks communicating via Universal Serial Bus (USB) interfaces, wireless networks (e.g., IEEE 802.11, IEEE 802.15, Bluetooth), networks communicating via in-premises power lines, and others. The lines connecting the interface **120** with the other devices in the premises **102a** may represent wired or wireless connections, as may be appropriate for the type of local network used. One or more of the devices at the premises **102a** may be configured to provide wireless communications channels (e.g., IEEE 802.11 channels) to communicate with one or more of the mobile devices **125**, which may be on- or off-premises.

[0029] The RFID tag reader **150** may be configured to interrogate or scan an RFID tag, such as RFID tag **160**. The RFID tag reader **150** may scan the RFID tag **160** by sending a signal (e.g., transmitting an RF signal). The RFID tag **160** may receive the signal sent by the RFID tag reader **150**. The RFID tag **160** may be configured to provide different data in response to the signal received from the RFID tag reader **150** based on the orientation of the RFID tag **160**. For example, when the RFID tag **160** is in a first orientation, the RFID tag **160** may provide first data, when the RFID tag **160** is in a second orientation, the RFID tag **160** may provide second data, and when the RFID tag **160** is in an Nth orientation, the RFID tag **160** may provide Nth data.

[0030] Any type of data may be provided by the RFID tag **160**. The data provided by the RFID tag **160** may be received by the RFID tag reader **150**. The data provided by the RFID tag **160** may be provided via a modulated signal based on the signal received from the RFID tag reader **150**. For example, the RFID tag **160** may send (e.g., transmit) a backscattered signal that may be modulated based on data that may be determined based on a particular orientation of the RFID tag **160**. The RFID tag reader **150** may receive the modulated backscattered signal from the RFID tag **160** and may determine the data provided by the RFID tag **160** by demodulating the modulated backscattered signal. The data provided by the RFID tag **160** may convey various different information to the RFID tag reader **150** including, for example, an ID of RFID tag **160** and/or an instruction or a command issued by the RFID tag **160**.

[0031] As a first example, when the RFID tag **160** is in a first orientation, the RFID tag **160** may provide data comprising a unique ID assigned to the RFID tag **160**. When the RFID tag is in all other orientations, the RFID tag **160** may provide no data (or may provide a fake or dummy ID, or may provide null data). In this manner, the RFID tag **160** may provide sensitive data—for example, its unique ID—only when in a certain orientation. In doing so, the RFID tag **160** may be prevented from providing its unique ID at times and only when a user chooses to have the RFID tag **160** do so. In turn, this reduces the risk of a bad actor covertly scanning the RFID tag **160** to obtain any sensitive data stored on the RFID tag **160**.

[0032] As a second example, when the RFID tag **160** is in a first orientation, the RFID tag **160** may provide data that may indicate authorization of a transaction. When the RFID tag is in all other orientations, the RFID tag **160** may provide no data (e.g., null data, or may provide an indication to not authorize a transaction). In this manner, the RFID tag **160** may be used as part of an MFA process to authorize a transaction only when in a certain orientation. In doing so, the RFID tag **160** may be prevented from authorizing a transaction at all times and only when a user chooses to have the RFID tag **160** do so.

[0033] As a third example, when the RFID tag **160** is in a first orientation, the RFID tag **160** may provide data that may indicate an alert or emergency signal. When the RFID tag is in all other orientations, the RFID tag **160** may provide no data (e.g., null data, or may provide an indication that there is no emergency or alert). In this manner, the RFID tag **160** may be used to signal an alarm condition (e.g., either overtly or covertly).

[0034] As a fourth example, the RFID tag **160** may be used to determine an orientation of a shipping container (e.g., a shipping box associated with the RFID tag **160** and used to transport a product). The RFID tag **160** may provide different data for each different possible orientation of the shipping container (e.g., each of six different orientations of the shipping container may cause the RFID tag **160** to provide one of six different data signals) or the RFID tag **160** may provide the same data for all acceptable orientations of the shipping container and may provide different data for all unacceptable orientations of the shipping container. In this manner, a determination as to whether the shipping container is in an acceptable orientation (e.g., the shipping container is right side up) or is in an unacceptable orientation (e.g., the shipping container has fallen over), may be quickly and easily made by scanning an area (e.g., the back of a shipping truck) containing the shipping container (e.g., the RFID tag **160** may be attached to the shipping container). This approach may be extended to determine how many shipping containers in an entire shipping truck are oriented incorrectly—for example, by scanning all of the shipping containers and paired RFID tags to determine which RFID tags return data indicating that they are oriented improperly. Unique IDs and/or unique data provided by each RFID tag in response to such scanning may allow identification of the exact shipping containers that are oriented incorrectly and therefore may be damaged.

[0035] The RFID tag reader **150** may process and/or interpret data received from the RFID tag **160** as any type of intended information such as, for example, a unique ID, an alert, a command, and/or a request to perform a certain function. Information from the RFID tag **160** may be provided via a single data signal or as a collection or sequence of data signals (e.g., such that a sequence of data provided by the RFID tag **160** is to be interpreted as providing a unique ID, an alert, a command, and/or a request to perform a certain function). The RFID tag reader **150** may receive any data provided by the RFID tag **160** and may provide the received data to any other device or component (e.g., any other device shown in FIG. 1). In doing so, any data provided by the RFID tag **160** may be used to initiate, cause, or terminate any action or function to be performed by any other device (e.g., any other device depicted in FIG. 1 or any other device communicatively coupled to the RFID tag reader **150**).

[0036] The mobile devices **125**, one or more of the devices in the premises **102a**, and/or other devices may receive, store, output, process, and/or otherwise use data associated with content items. A content item may comprise a video, a game, one or more images, software, audio, text, webpage(s), and/or other type of content. One or more types of data may be associated with a content item. A content item may, for example, be associated with media data (e.g., data encoding video, audio, and/or images) that may be processed to cause output of the content item via a display screen, a speaker, and/or other output device component.

[0037] As described above, one or more servers may be communicatively coupled to the external network **109**. For example, an RFID tag monitoring server **140** may communicate with the local office **103** (and/or one or more other local offices), one or more premises **102**, one or more access points **127**, one or more mobile devices **125**, and/or one or more other computing devices via the external network **109** (including, for example, the RFID tag reader **150**). The RFID tag monitor server **140** may monitor data received from the RFID tag **160**. For example, data provided by the RFID tag **160** may be received by the RFID tag reader **150**, and the RFID tag reader **150** may provide the received data to the RFID tag monitor server **140**. The RFID tag monitor server **140** may also enable the mobile devices **125** to receive data from the RFID tag **160** if the mobile devices **125** are located remote from the premises **102a**. The RFID tag monitor server **140** may

facilitate performance of any action, request, or command based on any data received from the RFID tag **150**.

[0038] Also or alternatively, the RFID tag monitor server **140** may be located in the local office **103**, in a premises **102**, and/or elsewhere in a network. The RFID tag monitor server **140** may communicate with an RFID tag monitor database **141**. The RFID tag monitor database **141** may store data received from the RFID tag **160**, may store information regarding the RFID tag **160** (e.g., an associated user, a unique ID of the RFID tag, etc.), and/or may store information related to events associated with the RFID tag **160**. For example, the RFID tag monitor database **141** may store information indicating a time of day and/or a location where the RFID tag **160** was used to authorize a financial transaction. Although shown as a separate element, the RFID tag monitor database **141** may be part of the RFID tag monitor server **140**. Also or alternatively, the push server **105**, the content server **106**, the application server **107**, the RFID tag monitor server **140**, and/or other server(s) may be combined. The servers **105**, **106**, **107**, and **140**, other servers, and/or the RFID tag monitor database **141** may be computing devices and may comprise memory storing data and also storing computer executable instructions that, when executed by one or more processors, cause the server(s) to perform steps described herein.

[0039] Any component depicted and described in relation to FIG. **1**—including, for example, the RFID tag reader **150**, the RFID tag monitor server **140**, and/or the RFID monitor database **141**—may receive data from the RFID tag **160** and may cause any operation or function indicated by the data or associated with the data to be performed, as further described herein.

[0040] FIG. **2** shows hardware elements of a computing device **200** that may be used to implement any of the computing devices shown or described in relation to FIG. **1** (e.g., the mobile devices **125**, any of the devices shown in the premises **102a**, any of the devices shown in the local office **103**, any of the wireless access points **127**, the RFID monitor server **140**, the RFID monitor database **141**, the RFID tag reader **150**, the RFID tag **160**, any devices that are part of or associated with the external network **109**) and any other computing devices discussed herein (e.g., a remote control unit associated with the gateway **111** and/or with another computing device). The computing device **200** may comprise one or more processors **201**, which may execute instructions of a computer program to perform any of the functions described herein. The instructions may be stored in a non-rewritable memory **202** such as a read-only memory (ROM), a rewritable memory **203** such as random-access memory (RAM) and/or flash memory, removable media **204** (e.g., a USB drive, a compact disk (CD), a digital versatile disk (DVD)), and/or in any other type of computer-readable storage medium or memory. Instructions may also be stored in an attached (or internal) hard drive **205** or other types of storage media. The computing device **200** may comprise one or more output components, such as a display device **206** (e.g., an external television and/or other external or internal display device) and a speaker **214**, and may comprise one or more output device controllers **207**, such as a video processor or a controller for an infra-red or BLUETOOTH transceiver. One or more user input devices **208** may comprise a remote control, a keyboard, a mouse, a touch screen (which may be integrated with the display device **206**), a microphone, etc. The computing device **200** may, for example receive sounds of speech input via a microphone. The processor **201** may (e.g., using one or more analog-to-digital (A/D) converters, digital signal processors (DSPs), and/or other components) digitize and/or otherwise generate audio data that is representative of the speech input. Also or alternatively, the computing device may comprise (e.g., in addition to the processor **201**) one or more A/D converters, DSPs, and/or other components that generate audio data that is representative of the speech input. The processor **201** and/or other components of the computing device may send speech data to one or more other computing devices, may receive (e.g., via network input/output (I/O) interface **210**, described below) speech data generated by another computing device, may perform speech recognition processing of speech data, and/or may perform other operations associated with speech data.

[0041] The computing device **200** may also comprise one or more network interfaces, such as the

network I/O interface **210** (e.g., a network card), to communicate with an external network **209**. The network I/O interface **210** may be a wired interface (e.g., electrical, RF (via coax), optical (via fiber)), a wireless interface, or a combination of the two. The network I/O interface **210** may comprise a modem configured to communicate via the external network **209**. The external network **209** may comprise the communication links **101** discussed above, the external network **109**, an in-home network, a network provider's wireless, coaxial, fiber, or hybrid fiber/coaxial distribution system (e.g., a DOCSIS network), or any other desired network. The computing device **200** may comprise a location-detecting device, such as a global positioning system (GPS) microprocessor **211**, which may be configured to receive and process global positioning signals and determine, with possible assistance from an external server and antenna, a geographic position of the computing device **200**.

[0042] Although FIG. **2** shows an example hardware configuration, one or more of the elements of the computing device **200** may be implemented as software or a combination of hardware and software. Modifications may be made to add, remove, combine, divide, etc. components of the computing device **200**. Additionally, the elements shown in FIG. **2** may be implemented using basic computing devices and components that have been configured to perform operations such as are described herein. For example, a memory of the computing device **200** may store computer-executable instructions that, when executed by the processor **201** and/or one or more other processors of the computing device **200**, cause the computing device **200** to perform one, some, or all of the operations described herein. Such memory and processor(s) may also be implemented through one or more Integrated Circuits (ICs). An IC may be, for example, a microprocessor that accesses programming instructions or other data stored in a ROM and/or hardwired into the IC. For example, an IC may comprise an Application Specific Integrated Circuit (ASIC) having gates and/or other logic dedicated to the calculations and other operations described herein. An IC may perform some operations based on execution of programming instructions read from ROM or RAM, with other operations hardwired into gates or other logic. Further, an IC may be configured to output image data to a display buffer.

[0043] FIG. **3** shows an RFID tag **300** that may provide different data or information based on corresponding different orientations of the RFID tag **300**. The RFID tag **300** may be an implementation of the RFID tag **160** depicted in FIG. **1**. The RFID tag **300** may be considered to be an RFID chip. The RFID tag **300** may be any type of RFID tag including a passive RFID tag, an active RFID tag, or a semi-passive RFID tag. The RFID tag **300** may be attached or embedded in any device or component that is capable of varying its orientation. As an example, the RFID tag **300** may be attached to any type of card such as, but not limited to, a credit card, a debit card, or an identification (ID) card. As another example, the RFID tag may be embedded into a person (e.g., into a finger of a person's hand or into the back of a person's hand) or into a pet or other animal.

[0044] For purposes of discussion only, the RFID tag **300** is depicted in FIG. **3** as a passive RFID tag but is not so limited. The RFID tag **300** may comprise an antenna **302**, a power circuit or component **304**, a tilt switch or orientation sensor **306**, a controller **308**, and a memory component **310**.

[0045] The antenna **302** may receive an RF signal from an RFID tag reader or interrogator (e.g., the RFID tag reader **150** depicted in FIG. **1**; not shown in FIG. **3** for simplicity). The RF signal received by the antenna **302** from an RFID tag reader may be considered to be an interrogation signal or a scanning signal (or input RF signal). The antenna **302** may transmit an RF signal in response to a received RF signal from an RFID tag reader. The RF signal transmitted by the antenna **302** in response to an RF signal received from an RFID tag reader may be considered to be a backscattered RF signal (or output RF signal).

[0046] The RF signal transmitted by the antenna **302** may be a modulated signal. The modulated signal may convey data or information. The information conveyed by the modulated signal, and therefore the modulated signal itself, may vary based on an orientation of the RFID tag **300**. For

example, in response to an RF interrogation signal, the RFID tag **300** may transmit (or send) a first modulation signal conveying first information when the RFID tag **300** is in a first orientation and may transmit a second modulation signal conveying second information when the RFID tag **300** is in a second, different orientation. In an alternative, the RFID tag **300** may include a first antenna (e.g., the antenna **302**) for receiving RF signals and a second antenna for transmitting RF signals. [0047] The power circuit **304** may extract or harvest power from an RF signal received by the antenna **302**. The power circuit **304** may comprise any combination of circuit elements or logic for generating power (e.g., DC power) from an RF signal received by the antenna **302**. For example, the power circuit **304** may comprise a rectifier or other circuit elements such as, but not limited to, a diode and/or a capacitor. The power circuit **304** may provide power to any other component of the RFID tag **300** such as, for example, the controller **308** and/or the memory component **310**.

[0048] The tilt switch **306** may be a switch or other logic or circuit component that is capable of sensing, detecting, and/or responding to different orientations of the RFID tag **300**. As an example, the tilt switch **306** may be configured to sense a first orientation of the RFID tag **300** and may be configured to sense a second, different orientation of the RFID tag **300**. Continuing with this example, the tilt switch **306** may be in a first state (e.g., may provide an open circuit) when the RFID tag **300** is in a first orientation and may be in a second state (e.g., may provide a closed circuit) when the RFID tag **300** is in a second orientation. As another example, the tilt switch **306** may be configured to be in a first state (e.g., may provide an open circuit) in all orientations of the RFID tag **300** other than a predetermined orientation of the RFID tag, in which the RFID tag may be in a second state (e.g., may provide a closed circuit).

[0049] The state of the tilt switch **306** may be detected or sensed by the controller **308**. For example, the tilt switch **306** may provide a signal to the controller **308** that may be used by the controller **308** to determine a state of the tilt switch **306**. As another example, the controller **308** may detect an orientation of the RFID tag **300** by detecting a state of the tilt switch **306**-by detecting whether the tilt switch is providing a closed circuit or an open circuit. Overall, the controller **308** may be configured to determine an orientation of the RFID tag **300** based on a state of the tilt switch **306**.

[0050] The controller **308** may be any type of controller including any type of microchip, processor, microcontroller, logic, or digital signal processor (DSP), or any combination thereof. The controller **308** may control operation of the RFID tag **300**. The controller **308** may receive power from the power circuit **304**. As an alternative, the RFID tag **300** may include a power source (e.g., a battery) that may supply power to the controller **308** alone or in conjunction with the power circuit **304**.

[0051] As discussed above, the controller **308** may detect a state of the tilt switch **306** and therefore detect an orientation of the RFID tag **300**. Any number of orientations of the RFID tag **300** may be detected by the controller **308** based on a corresponding number of states of the tilt switch **306**. As an example, the controller **308** may detect whether the RFID tag **300** is positioned in a first orientation (e.g., based on the tilt switch providing an open circuit or a first signal) and may detect whether the RFID tag is positioned in a second orientation (e.g., based on the tilt switch providing a closed circuit or a second, different signal).

[0052] The controller **308** may be coupled to the memory component **310**. The memory component **310** may comprise any number of memory components and may comprise any type of memory (e.g., any type of non-volatile memory). The memory component **310** may store one or more information signals, data, or information sequences, corresponding to one or more orientations of the RFID tag **300**, that may be used to generate one or more corresponding output or backscattered RF signals.

[0053] As an example operation of the RFID tag **300**, the controller **308** may read out different information sequences or signals stored in the memory component **310** based on a detected state of the tilt switch **306**. For example, when the RFID tag is positioned in a first orientation, the tilt

switch **306** may be in a first state, and a first information sequence may be read out from the memory component **310**. The first information sequence may comprise binary data. The first information sequence may be used to modulate a received RF signal **312** to generate a first output or backscattered RF signal **314**. The first output RF signal **314** may convey any first information (e.g., a first ID, a first command, a first request, etc.).

[0054] When the RFID tag **300** is positioned in a second orientation, the tilt switch **306** may be in a second state, and a second information sequence may be read out from the memory component **310**. The second information sequence may be used to modulate the received RF signal **312** to generate a second output or backscattered RF signal **316**. The second output RF signal may convey any second information (e.g., a second ID, a second command, a second request, etc.). As an alternative, the RFID tag **300** may be configured to transmit an output RF signal when the RFID tag **300** is in a predetermined orientation and not to transmit any RF signal when the RFID tag **300** is in any other orientation (or may transmit null data). Accordingly, the RFID tag **300** may transmit a first output RF (e.g., the first output RF signal **314**) when the RFID tag **300** is in a first specific orientation or is oriented within a first range of orientations (e.g., corresponding to a first state of an orientation sensor), and the RFID tag **300** may transmit a second output RF (e.g., the second output RF signal **3146**) when the RFID tag **300** is in a second specific orientation or is oriented within a second range of orientations (e.g., corresponding to a second state of an orientation sensor).

[0055] The controller **308** may control modulation of any output RF signal based on any information sequence stored by the memory component **310**. The controller **308** may modulate any output RF signal according to any modulation scheme or technique including, but not limited to amplitude-shift keying (ASK), phase-shift keying (PSK), on-off keying (OOK), etc.

[0056] As shown in FIG. **3**, the first output RF signal **314** may be transmitted in response to the received RF signal **312**, which may be considered an interrogation signal that is transmitted by an RFID tag reader. As additionally shown in FIG. **3**, the second output RF signal **316** may be transmitted in response to the received RF signal **312**. As an alternative, the RFID tag **300** may transmit the first output RF signal **314** when the RFID tag **300** is in a predetermined orientation and may not transmit any RF signal when the RFID tag **300** is in any other orientation (e.g., such that the second RF output signal **316** is not transmitted).

[0057] Also, or alternatively, one of a plurality of output RF signals may be provided based on a plurality of corresponding orientations of the RFID tag **300**, as sensed by the tilt switch **306**. For example, the tilt switch **306** may be capable of detecting N different orientations of the RFID tag **300** and the memory component **310** may store N corresponding different information sequences (e.g., where N is an integer). Based on the tilt switch **306** detecting one of the N different orientations of the RFID tag **300**, the controller **308** may read out one of the N different information sequences stored by the memory component **310** that corresponds to the detected orientation of the RFID tag **300**. The controller **308** may then cause the antenna **302** to transmit one of N different output RF signals based on the determined information sequence. In this manner, any number of different output RF signals may be transmitted by the RFID tag **300** based on a corresponding number of different orientations of the RFID tag **300** being sensed by the tilt switch **306**, thereby allowing the RFID tag **300** to provide or convey any number of different IDs, commands, requests, etc. A single tilt switch may be configured to detect N different orientations to the RFID tag **300** or any number of tilt switches may be used together to detect the N different orientations to the RFID tag **300**.

[0058] Any RF output signal transmitted by the RFID tag **300** may be modulated by shunting the antenna **302**, by varying any properties of the antenna **302**, and/or by varying any properties of any circuit or circuit element of the RFID tag **300** that may be associated with transmitting an RF output signal. As an example, the orientation of the RFID tag **300** may cause the tilt switch **306** to add or remove various different circuit elements that may be associated with transmitting an RF output signal, thereby allowing properties of the RF output signal to vary based on the

properties of the circuit elements added or removed. In general, any modulation technique implemented by RFID tags may be used by the RFID tag **300**.

[0059] FIG. **4A** shows an example tilt switch **400** in a first position. The tilt switch **400** may be an implementation of the tilt switch **306**. The first position of the tilt switch **400** depicted in FIG. **4A** may correspond to a closed state or a first state of the tilt switch **400**.

[0060] The tilt switch **400** may comprise a case **402**, an outer pin or connector **404**, a central pin or connector **406**, a ball bearing **408**, and a frame **410**. The case **402** may comprise a conductive material such as metal. The case **402** may house the ball bearing **408** within an inner (open) region of the case **402**. The ball bearing **408** may also comprise a conductive material such as metal. The inner (open) region of the case **402** may be filled with a non-conductive liquid (e.g., an oil) to dampen/slow movement of the ball bearing **408** (e.g., to prevent rattling).

[0061] The case **402** may be supported by the frame **410** (e.g., connected or attached to the frame **410**). The frame **410** may comprise a non-conductive material. The central pin **406** may be supported by and may extend through the frame **410** and into the inner region of the case **402** that houses the ball bearing **408**. As shown in FIG. **4A**, the central pin **406** may be positioned centrally with respect to the frame **410** and the case **402**. The outer pin **404** may also be supported by the frame **410** and may be positioned peripherally with respect to the frame **410**. The outer pin **404** may be connected to an outer portion of the case **402**. The outer pin **404** and the central pin **406** may both comprise a conductive material such as metal. The central pin **406** may be electrically insulated from the case **402**. The outer pin **404** may be electrically connected to the case **402**.

[0062] The tilt switch **400** may operate by detecting or sensing a roll or movement of the ball bearing **408** within the case **402**. The ball bearing **408** may be allowed to move or roll around within the inner region of the case **402**—for example, as an RFID tag containing the tilt switch **400** is moved. Based on a position or tilt of the tilt switch **400**, the ball bearing **408** may at times contact central pin **406**. As shown in FIG. **4A**, when the ball bearing **408** contacts the central pin **406**, the central pin **406**, the ball bearing **408**, the case **402**, and the outer pin **404** may together be electrically connected, thereby completing (or closing) a circuit that may indicate or represent the tilt switch **400** being in a closed position (or a first position/first state). The electrical connectivity between the central pin **406**, the ball bearing **408**, the case **402**, and the outer pin **404** as depicted in FIG. **4A** may be detected or sensed by a controller (e.g., the controller **308**) of an RFID tag (e.g., the RFID tag **300**).

[0063] FIG. **4B** shows the tilt switch **400** in a second position. The second position of the tilt switch **400** depicted in FIG. **4B** may correspond to an open state or a second state of the tilt switch **400**. As shown in FIG. **4B**, the ball bearing **408** may no longer contact the central pin **406**. As a result, the central pin **406**, the ball bearing **408**, the case **402**, and the outer pin **404** may no longer be electrically connected together, thereby not completing (or opening) a circuit that may indicate or represent the tilt switch **400** being in an open position (or a second position/second state). The lack of electrical connectivity between the central pin **406**, the ball bearing **408**, the case **402**, and the outer pin **404** as depicted in FIG. **4B** may be detected or sensed by a controller (e.g., the controller **308**) of an RFID tag (e.g., the RFID tag **300**).

[0064] As described above, whether the ball bearing **408** contacts the central pin **406** or not—and therefore completes a circuit or not, respectively—may be detected by the controller **308**. As explained herein, when the controller **308** detects that the tilt switch **400** is in a closed position (e.g., as shown in FIG. **4A**), the controller **308** may cause the RFID tag **300** to transmit a first RF output signal. When the controller **308** detects that the tilt switch **400** is in an open position (e.g., as shown in FIG. **4B**), the controller **308** may cause the RFID tag **300** to transmit a second RF output signal. Each RF output signal may then be received and processed to determine what information was conveyed (e.g., which ID, which command, which request, etc.).

[0065] The tilt switch **306** is not limited to the implementation of the tilt switch **400** shown in FIGS. **4A** and **4B**. Any type of tilt switch that is capable of sensing one or more different positions

or orientations may be used by the RFID tag **300**. For example, a tilt switch (e.g., the tilt switch **400**) may include multiple pins positioned at various angles that may come into contact with a moveable conductive member (e.g., the ball bearing **408**) when the tilt switch is held in different positions, thereby enabling the tilt switch to sense various orientations (i.e., more than two orientations). Further, multiple different tilt switches may be used in combination—for example, positioned at various orientations and/or including various pins arranged in various orientations—to detect multiple different orientations of an RFID tag. As another example, a tilt switch may include a liquid conductive material (e.g., mercury) in lieu of a solid conductive member (e.g., instead of the ball bearing **408**) that may be capable of contacting differently oriented pins arranged in an open inner region (e.g., the inner region of the case **402**) of the tilt switch, thereby allowing multiple different orientations to be detected. Accordingly, one or more tilt switches may be arranged and communicatively coupled so as to be able to detect three or more distinct orientations. [0066] FIG. 5 shows an example range of orientations of an example tilt switch **512** when in a closed position (e.g., a first state) and when in an open position (e.g., a second state). The tilt switch **512** may provide similar functionality as the tilt switch **400** in that it may detect two different orientation states. FIG. 5 includes a reference indicator **502** and a reference orientation sphere **504**. The reference indicator **502** may be perpendicular to a top outer surface of the tilt switch **512**. The reference orientation sphere **504** may indicate a range of different positions or orientations of the tilt switch **512** relative to the reference indicator **502**. Over a first range of orientations **506** (e.g., relative to a direction the reference indicator **502** is pointing), the tilt switch **512** may be in the open position. Over a second range of orientations **508** (e.g., relative to a direction the reference indicator **502** is pointing), the tilt switch **512** may be in the closed position. When the reference indicator **502** is pointing directly upwards (e.g., relative to a zero-degree reference of the reference orientation sphere **504**), a state of the tilt switch **512** may be similar to the orientation of the tilt switch **400** as shown in FIG. 4B. When the reference indicator is pointing directly downwards (e.g., relative to the zero-degree reference of the reference orientation sphere **504**), a state of the tilt switch **400** may be similar to the orientation of the tilt switch **400** as shown in FIG. 4A.

[0067] As an example operation of the tilt switch **512**, the tilt switch **512** may initially be in a closed position. As the tilt switch **512** is moved or tilted, the tilt switch **512** may progress to an open position. A tilt angle **510** may indicate an amount of movement or tilt of the tilt switch **512** that may cause the tilt switch **512** to switch from the closed position to the open position. The tilt angle **510** may correspond to an angle of the reference indicator **502** relative to a top of the tilt switch **512** when the tilt switch **512** is orientated such that the reference indicator **502** is pointing directly upwards (e.g., relative to a plane perpendicular to the reference indicator **502** when it is pointing directly upwards). The tilt angle **510** provides some tolerance for allowing the tilt switch **512** to remain in a closed state before transitioning to an open state (e.g., such that the tile switch **512** does not need to be held exactly right side up to remain in the closed state). As described herein, the tilt switch **512** may be configured to transmit a first RF output signal when oriented in any position within the first range of orientations **506** and to transmit a second RF output signal when oriented in any position within the second range of orientations **508**. The tilt switch **512** may be used in combination with, or instead of, any other orientation sensor described herein including, for example, the tilt switch **400**.

[0068] FIG. 6 shows different output signals that may be provided by the RFID tag **300** based on different corresponding orientations of the RFID tag **300**. FIG. 6 includes an RFID tag reader or integrator **602**, an RFID tag **300-1** in a first orientation, and an RFID tag **300-2** in a second orientation. The RFID tags **300-1** and **300-2** may represent the RFID tag **300** in different orientations. The RFID tag reader **602** may represent an implementation of the RFID tag reader **150** depicted in FIG. 1.

[0069] The RFID tag reader **602** may transmit a signal **604**. The signal **604** may be an RF signal

and may be considered to be an interrogation or scanning signal. As an example, the signal **604** may be a continuous waveform. Reference indicator **606** may indicate an orientation of the RFID tag **300-1**. For example, the reference indicator **606** may indicate that the RFID tag **300-1** is in a first orientation that may correspond to an open position of the tilt switch **306**. Reference indicator **608** may indicate an orientation of the RFID tag **300-2**. For example, the reference indicator **608** may indicate that the RFID tag **300-2** is in a second orientation that may correspond to a closed position of the tilt switch **306**.

[0070] The RFID tag **300-1**, oriented in the first position, may receive the signal **604**. Reception of the signal **604** may cause the RFID tag **300-1** to read out a first information sequence **610** from the memory **310**. As an example, the first information sequence **610** may be a binary sequence such as 110011. Reception of the signal **604** may further cause the RFID tag **300-1** to output signal **612** based on the first information sequence **610** and the signal **604**. For example, the output signal **612** may be a backscattered modulated signal that is a modulated version of the signal **604**. The output signal **612** may comprise a version of the signal **604** (e.g., a reflected and/or attenuated version of the signal **604**) modulated by the first information sequence **610**. The modulated output signal **612** may convey first information from the RFID tag **300-1** to the RFID tag reader **602**. The RFID tag reader **602** may interpret this first information, for example, as a first command or a first ID.

[0071] The RFID tag **300-2**, oriented in the second position, may also receive the signal **604**. Reception of the signal **602** may cause the RFID tag **300-2** to read out a second information sequence **614** from the memory **310**. As an example, the second information sequence **614** may be a binary sequence such as 001011. Reception of the signal **604** may further cause the RFID tag **300-2** to output signal **616** based on the second information sequence **614** and the signal **604**. For example, the output signal **616** may be a backscattered modulated signal that is a modulated version of the signal **604**. The output signal **616** may comprise a version of the signal **604** (e.g., a reflected and/or attenuated version of the signal **604**) modulated by the second information sequence **614**. The modulated output signal **616** may convey second information from the RFID tag **300-2** to the RFID reader **602**. The RFID reader **602** may interpret this second information, for example, as a second command or a second ID.

[0072] In this manner, different orientations of the RFID tag **300**, as detected by the tilt switch **306**, may cause the RFID tag **300** to transmit different RF signals to the RFID tag reader **602** based on a signal **604** initially transmitted by the RFID tag reader **602**. By transmitting different RF signals to the RFID tag reader **602** based on corresponding different orientations of the RFID tag **300**, the RFID tag **300** may provide different information to the RFID tag reader **602** based on the orientation of the RFID tag **300** (e.g., at the time of receiving and/or processing the signal **604**). This enables any type of command or ID, or any type of sequence of commands or IDs, to be conveyed to the RFID tag reader **602** by the RFID tag **300**. In turn, as explained in more detail below, the RFID tag reader **602** may interpret the received commands and/or IDs from the RFID tag **300** to facilitate performance of any function such as, for example, authorize a transaction or identify an individual. Overall, the RFID tag **300** enables different information to be provided by the RFID tag **300** based on its orientation, and the different information that may be provided by the RFID tag **300** may be used by the RFID tag reader **602** to trigger performance of any function.

[0073] FIG. 7 shows an RFID tag **700** that may provide different information based on corresponding different orientations of the RFID tag **700** that may also include various additional feature detection switches. The RFID tag **700** may include various components that are similar to or the same as the components of the RFID tag **300**. The RFID tag **700** may also operate in a similar or the same manner as the RFID tag **300** as described herein. The RFID tag **700** may be an implementation of the RFID tag **160** depicted in FIG. 1.

[0074] FIG. 7 shows that the RFID tag **700** may additionally include a temperature detection switch **702**. The temperature detection switch **702** may include any type of thermal detection component capable of changing states (e.g., from a first state to a second state) based on a detected

temperature. For example, the temperature detection switch **702** may be in an open state (e.g., a first state) when the temperature detection switch **702** detects a temperature that satisfies a predetermined temperature threshold (e.g., when the detected temperature meets or exceeds the predetermined temperature threshold). As a further example, the temperature detection switch **702** may be in an open state (e.g., a first state) when the temperature detection switch **702** detects a temperature that exceeds 90 degrees Fahrenheit.

[0075] The temperature detection switch **702** may be in a closed state (e.g., a second state) when the temperature detection switch **702** detects a temperature that does not satisfy a predetermined temperature threshold (e.g., when the detected temperature does not meet or exceed the predetermined temperature threshold). As a further example, the temperature detection switch **702** may be in a closed state (e.g., a second state) when the temperature detection switch **702** detects a temperature that is below 90 degrees Fahrenheit.

[0076] The state of the temperature detection switch **702** may be indicated by the RFID tag **700** to an RFID tag reader (e.g., the RFID tag reader **150** depicted in FIG. **1**). As an example, the state of the temperature detection switch **702** may change which information sequence is read from the memory **310** and used to modulate a transmitted RF output signal by the RFID tag **700**. By doing so, the RFID tag **700** may provide information indicating a detected temperature in relation to the predetermined temperature threshold. For example, when the temperature detection switch **702** detects a first temperature associated with the RFID tag **700** that is less than a predetermined temperature threshold, the RFID tag **700** may transmit a first RF output signal based on a first stored information sequence when the RFID tag **700** is in a first orientation. When the temperature detection switch **702** detects a second temperature associated with the RFID tag **700** that is greater than the predetermined temperature threshold, the RFID tag **700** may transmit a second RF output signal based on a second stored information sequence when the RFID tag **700** is in the first orientation. Accordingly, the RFID tag **700** may transmit different signals and may convey different information (e.g., different information about a detected temperature associated with the RFID tag **700**) when the RFID tag is in the same orientation but is associated with a different detected temperature.

[0077] The detected temperature may be any temperature of any device or thing. For example, the RFID tag **700** may be embedded in a human and may indicate, based on the state of the RFID tag, whether the RFID tag **700** is embedded in a person or has been removed from the person. If the RFID tag **700** indicates that it has been removed from the person (e.g., because a temperature threshold is not met), then certain RF output signals that may be transmitted by the RFID tag **700** may no longer be valid, or may no longer even be capable of being transmitted. This may prevent an RF output signal from being transmitted that may otherwise indicate authorization for a command or operation (e.g., for a financial transaction), thereby providing enhanced authorization security.

[0078] The temperature detection switch **702** may be capable of transitioning between different states based on a repeated comparison of a detected temperature to a predetermined temperature threshold, or the temperature detection switch **702** may transition from a first state to a second state permanently once a certain predetermined temperature threshold is crossed. The RFID tag **700** may include any other types of environmental detection switches or components that may change or alter information that may be provided by the RFID tag **700** based on a state of any environmental detection switch. As an example, the RFID tag **700** may include a shock detection switch (which may be considered to be a shock sensor switch). A shock detection switch may permanently switch from a first state to a second state when a shock or force of a certain amount is absorbed by the RFID tag **700** (or a device or component connected or in close proximity to the RFID tag **700**). For example, the RFID tag **700** may be connected to a television. A shock detection switch of the RFID tag **700** may absorb a force when the television is dropped during shipping. As a result, the shock detection switch may permanently switch from a first state (e.g., an open state) to a second state

(e.g., a closed state) to indicate that a force that satisfies a predetermined force threshold has been absorbed. As explained above, the state of the shock detection switch may influence the information provided by the RFID tag **700** such that an indication of the absorbed force may be conveyed to an RFID tag reader. Any type of environmental detection switch may be included in the RFID tag **700**, with each environmental detection switch capable of operating within any number of states to thereby alter any information provided by the RFID tag **700** in any number of corresponding manners. Other detection switches or sensors may be included within RFID tag **700** such as, for example, switches or sensors that detect humidity, magnetic flux, or gas, and the RFID tag **700** may provide different data for a given orientation based on a state of any additional switch or sensor.

[0079] FIG. **8** shows an RFID tag reader **800**. The RFID tag reader **800** may be an implementation of the RFID tag reader **150** depicted in FIG. **1**. The RFID tag reader **800** may comprise a lower portion or pad **802**, a stand **804**, and a magnet **806**. The pad **802** may contain components to enable the RFID tag reader **800** to transmit an RF output signal (e.g., an RFID tag interrogation signal) and to receive an RF input signal (e.g., an RF signal outputted by an RFID tag). As an example, a person's finger may contain an embedded RFID tag (e.g., the RFID tag **300** or the RFID tag **700**). The person may position the person's hand over the pad **802** in order for the embedded RFID tag to be interrogated or read by the RFID reader **800**. An open area over the pad **802** enables the person to orient the person's hand in various positions to cause the embedded RFID tag to transmit any number of RF output signals, based on the person's hand orientation (and therefore orientation of the embedded RFID tag).

[0080] The RFID tag reader **800** may be configured to receive any waveform transmitted by an RFID tag. The RFID tag reader **800** may be configured to process any waveform or series of waveforms to receive any information (e.g., a code, an ID, a command, etc.) intended to be conveyed via any waveform or series of waveforms transmitted by an RFID tag read by the RFID tag reader **800**. The RFID tag reader **800** may store and/or transmit any such received information to any other computing device that may be coupled to the RFID tag reader **800** (e.g., any other device or component depicted in FIG. **1**).

[0081] The stand **804** may be connected to the pad **802**. The stand **804** may house the magnet **806**. The magnet **806** may be selectively activated to invoke a certain state of a tilt switch contained with an RFID tag (e.g., the tilt switch **306**). As an example, the magnet **806** may typically or usually be deactivated such that it does not influence the operation of a tilt switch contained within an RFID tag (e.g., the RFID tag **300** or the RFID tag **700**). However, at times it may be desirable to activate the magnet **806** to attract a metal component of a tilt switch (e.g., the ball bearing **408**). As a result, the metal component of the tilt switch may be attracted upwards towards the magnet **806** relative to the pad **802**, as indicated by direction indicator **808** (e.g., in a direction orthogonal to a plane of the pad **802**). In doing so, the RFID tag reader **800** may cause the tilt switch to transition to a certain state (e.g., an open or closed state) based on a movement of the metal component of the tilt switch.

[0082] The RFID tag reader **800** may comprise a user interface (not depicted in FIG. **8** for simplicity) allowing a user to control operation of the RFID tag reader **800**. The RFID tag reader **800** may be coupled to a network and may be coupled to one or more other computing devices including, but not limited to, a remote computing device or database, as described above in relation to FIG. **1**.

[0083] FIG. **9** shows example hand orientations that may correspond to different orientations of an RFID tag (e.g., the RFID tag **300** or the RFID tag **700**) that may be embedded in a person's hand or finger. For purposes of discussion only, FIG. **9** includes a first reference diagram **902** that shows three reference axes indicating different movements that are possible when an object (e.g., an airplane) moves within a space or medium (e.g., a three-dimensional space). Specifically, the first reference diagram **902** shows a first movement—roll—that may occur about a reference

longitudinal axis, a second movement—yaw—that may occur about a reference vertical axis, and a third movement—pitch—that may occur about a reference lateral axis, relative to the orientation of the object shown.

[0084] As with the object shown in the first reference diagram **902**, a second reference diagram **904** of FIG. **9** shows the three different movements that are possible for a person's hand relative to the reference axes. As shown in the second reference diagram **904**, a roll indicator **906**, a yaw indicator **908**, and a pitch indicator **910** show the various manners the person's hand may move or rotate relative to three axes of movement.

[0085] In view of the first reference diagram **902** and the second reference diagram **904**, FIG. **9** further shows a third reference diagram **912**. The third reference diagram **902** shows the movement of a person's hand as it may be rotated about the three different reference axes relative to the RFID tag reader **802**. Specifically, a top portion (top row) **914** of the third reference diagram **912** shows how the roll of a person's hand may be varied with respect to the RFID tag reader **802**. The middle portion (middle row) **916** of the third reference diagram **912** shows how the pitch of a person's hand may be varied with respect to the RFID tag reader **802**. The lower portion (lower row) **918** of the third reference diagram **912** shows how the yaw of a person's hand may be varied with respect to the RFID tag reader **802**.

[0086] When embedded in a person's hand, an RFID tag (e.g., the RFID tag **300** or the RFID tag **700**) may be configured to transmit a particular RF output signal based a corresponding roll, pitch, and yaw orientation. Based on the particular roll, pitch, and yaw of the person's hand, corresponding information may be provided to the RFID tag reader **800**. In this manner, various different roll, pitch, and yaw orientations of the person's hand may enable a person to issue various commands or provide various data to an RFID tag reader.

[0087] As an example, a specific combination of roll, pitch, and yaw orientations of a person's hand may cause an embedded RFID tag to transmit a code specific to the person (e.g., an authorization code or ID code), while all other orientations of the person's hand may not cause the embedded RFID tag to transmit the code specific to the person (e.g., a false code may be transmitted or no transmission may be made). This technique may prevent an eavesdropper from stealing the person's code passively without the person's knowledge. That is, an eavesdropper armed with an RFID tag reader would not be able to read the code from the RFID tag embedded in the person's hand unless the person were to position the person's hand in a specific orientation that allows the code to be read.

[0088] The orientations shown and described in relation to FIG. **9** are not limited to RFID tags embedded within a person's hand and may also or alternatively be used to determine what information is provided by an RFID tag that is provided in conjunction with any item such as, for example, a card (e.g., a credit card or ID card) that may contain an RFID tag as described herein.

[0089] FIG. **10** shows various example user RFID activation orientations. The RFID activation orientations may relate to different users (e.g., Users A, B, and C) that each have an RFID tag (e.g., the RFID tag **300** or the RFID tag reader **800**) embedded within their hands. Each embedded RFID tag may be associated with a unique code or ID for the users that may be used to provide an authorized command or instruction (e.g., to authorize a financial transaction).

[0090] As shown, a first orientation **1002** is associated with User A, a second orientation **1004** is associated with User B, and a third orientation **1006** is associated with User C. As a specific example, when User A positions User's A hand in an orientation corresponding to a roll of $30\pm5^\circ$, a pitch of $0\pm5^\circ$, and a yaw of $-5\pm5^\circ$, the RFID tag embedded in User A's hand may be configured to transmit an RF output signal that provides a code, activation signal, or other information intended to convey an authorized command, instruction, or other data to an RFID tag reader. As shown, the first orientation **1002** for User A allows for some slight variation in the orientation by allowing for a five-degree variance in each direction of movement (indicated by the " $\pm5^\circ$ " shown in the table).

[0091] The orientations shown in FIG. **10** may be considered to be active orientations because the

orientations may cause a valid or authorized data signal to be provided by the RFID tag. In all other orientations, for example, the RFID tag may not provide the valid or authorized data signal (e.g., no data signal may be provided or a dummy data signal may be provided). By designating an activate orientation for a user, the user may be assured that the RFID tag is only providing sensitive data (e.g., the valid or authorized data signal) when the user specifically positions the user's hand in a certain manner. In doing so, the likelihood that the RFID tag provides sensitive data at other times (e.g., when a user is holding the user's hand in another orientation such as when walking) is reduced, thereby preventing a bad actor from attempting to read sensitive data from the RFID tag without the user's knowledge. Active orientations may be customized to a user. Further, the activate orientations as described in relation to FIG. 10 are not limited to embedded RFID tags but are applicable to any device (e.g., a card) that contains or is connected to an RFID tag.

[0092] As described herein, different orientations of an embedded RFID tag may cause different corresponding data or codes to be transmitted. As such, a user with an embedded tag may be able to output various different data sequences that correspondingly cause different actions to occur based on a particular sequence of orientations of the user's hand. For example, an embedded RFID tag of a user may output a first code when the user holds his hand in a first orientation, may output a second code when the user holds his hand in a second orientation, and may output a third code when the user holds his hand in a third orientation. This may allow the user to issue a first command to cause a first action to occur when the user holds his hand in a first sequence of orientations—for example, a sequence of orientations that include the first orientation, the third orientation, and then the first orientation again such that the first code, the third code, and then first code again are provided in sequence. Further, a second command to cause a second action to occur may be issued in response to the user holding his hand in a second, different sequence of orientations—for example, a sequence of orientations that include the first orientation, the third orientation, and the second orientation such that the first code, the third code, and the second code are provided in sequence.

[0093] The table below summarizes various example use cases involving an RFID tag as described herein (e.g., the RFID tag 300 or the RFID tag reader 800). For use case no. 1, using an RFID tag as described herein to authorize a financial transaction may reduce risks associated with financial accounts as a bad actor would be unable to conduct fraudulent financial activity without access to and knowledge of a user's particular RFID orientation that results in authorization of a transaction. For use case no. 2, using an RFID tag as described herein to secure personal property (e.g., a bike, a car, etc.) may reduce the likelihood that such property is stolen as a bad actor would be unable to unlock or access the personal property without access to and knowledge of a user's particular RFID orientation that results in granting access to the property (e.g., unlocking a car door, unlocking a chain lock, etc.). For use case no. 3, using an RFID tag as described herein to provide an emergency alert may allow a user to either openly or covertly communicate an indication of an alarm condition to any remote system or individual. For example, a user may hold the RFID tag in a specific orientation that indicates that the user is in immediate danger and the RFID tag reader may provide that information to appropriate emergency response individuals and systems (e.g., the police). For use case no. 4, using an RFID tag as described herein to activate or deactivate a car based on the RFID tag being positioned close to a certain area of the car (e.g., the steering wheel) and/or in a specific orientation may prevent carjacking. For use case no. 5, using an RFID tag as described herein to control a device may allow for intuitive or fast responding control of the device. Any type of device may be controlled by the user. As an example, one or more RFID tags may be embedded in one or more fingers of a user, and certain orientations of the user's fingers may cause certain control of a device. The device may be caused to move or to operate in a certain manner based on the user's commands.

TABLE-US-00001 TABLE 1 Example RFID tag use case scenarios

Use Case No.	Use Case
Example Scenario 1	MFA of a financial Point of sale (POS) device may include an RFID tag

transaction. reader. User may use a financial card (e.g., a credit card) to finance a transaction. User may then use an RFID tag held in a specific orientation to authorize the transaction. If the RFID tag is not held in the correct orientation, then the transaction will not be authorized. 2 Preventing theft of A car may include an RFID tag reader and may be personal property. unlocked if the user's RFID tag is held in a specific orientation (possibly in combination with using a key). If the RFID tag is not held in the correct orientation, then the car will not unlock. 3 Emergency alert RFID tag may be positioned into a specific orientation (audible or silent) to provide an alert to an RFID tag reader. Any type of alert may be provided (e.g., indicating that someone is attempting to steal an RFID tag, that someone is in imminent physical danger, etc.). RFID tag reader may route the alert to appropriate responding systems. 4 Preventing carjacking. An interior of a car may include an RFID tag reader. The car may be activated only if the user's RFID tag is held in a specific orientation. A user's car may stop working if the user's RFID tag is not positioned close to the car's steering wheel for a specified period of time. 5 Control of a device A user may issue commands/instructions for (e.g., a fighter jet). controlling a device based on specific orientations of an embedded RFID tag. For example, a user may control a fighter jet to turn or to fire a weapon. The device to be controlled may include an RFID tag reader that responds or causes certain actions in response to different commands received from one or more RFID tags.

[0094] FIG. **11** is a flow chart showing steps of an example method of an RFID tag providing data based on an orientation of the RFID tag. One, some, or all steps of the example method of FIG. **11** may be performed by any of the devices described herein (e.g., RFID tag **300** or the RFID tag reader **800**). Also or alternatively, one, some, or all steps of the example method of FIG. **11** may be performed by one or more other devices, including any of the computing devices described herein. Steps of the example method of FIG. **11** may be omitted, performed in other orders, and/or otherwise modified, and/or one or more additional steps may be added.

[0095] The method may start in step **1102**. In step **1102**, an RFID tag may detect whether the RFID tag is receiving an RF input signal. The RF input signal may be a scanning signal or an interrogation signal transmitted by an RFID tag reader.

[0096] If the RFID tag is not receiving an input RF signal, step **1104** may be performed. In step **1104**, the RFID tag may wait until an input RF signal is received and may repeat step **1102** as shown.

[0097] If the RFID tag detects reception of an input RF signal, step **1106** may be performed. In step **1106**, an orientation of the RFID tag may be determined. The orientation of the RFID tag may be determined using an orientation sensor (e.g., the tilt switch **306**).

[0098] In step **1108**, based on the determined orientation of the RFID tag, the RFID tag may determine data or other information (e.g., a unique ID, a bit sequence, etc.) to provide in response to the input RF signal. The data may be stored in a memory of the RFID tag (e.g., the memory component **310**).

[0099] In step **1110**, the RFID tag may transmit an output RF signal. The output RF signal may be a backscattered signal based on the input RF signal. The output RF signal may provide the data determined in step **1108**. For example, the output RF signal may be modulated based on the data determined in step **1108**. After step **1110**, step **1102** may be performed—for example, via wait step **1104**—and the RFID tag may await another input RF signal.

[0100] FIG. **12** is a flow chart showing steps of an example method of an RFID tag reader receiving data from an RFID tag based on the orientation of the RFID tag. One, some, or all steps of the example method of FIG. **12** may be performed by any of the devices described herein (e.g., RFID tag reader **802**). Also or alternatively, one, some, or all steps of the example method of FIG. **11** may be performed by one or more other devices, including any of the computing devices described herein. Steps of the example method of FIG. **12** may be omitted, performed in other orders, and/or otherwise modified, and/or one or more additional steps may be added.

[0101] The method of FIG. **12** may be implemented by an RFID tag reader to receive a sequence of

data from an RFID tag. For example, the method of FIG. 12 may be implemented when the RFID tag reader may provide one or more interrogation signals to the RFID tag. In response, the RFID tag may provide one or more responsive signals, with each signal providing different data based on the orientation of the RFID tag. In this manner, the RFID tag may send first data in response to a first interrogation signal from the RFID tag reader and may send second data in response to a second interrogation signal from the RFID tag reader. The RFID tag may change orientations between receiving the two different interrogation signals such that the RFID tag provides different data in response to each interrogation signal. The RFID tag reader may receive the first data and the second data and may determine if a valid command or instruction corresponds to the first data and the second data (e.g., in combination). As an alternative to transmitting one or more interrogation signals, the RFID tag reader may provide a single interrogation signal for a longer period of time, allowing the RFID tag to change orientations while receiving the interrogation signal so that the data provided in response to the interrogation signal may be varied. In this manner, a sequence of data may be provided to an RFID tag reader.

[0102] The method may begin at step **1202**. In step **1202**, the RFID tag reader may transmit (e.g., send) an interrogation signal. The interrogation signal may be an RF signal. The interrogation signal may be a scanning signal. The interrogation signal may be transmitted at or in the direction of an RFID tag. The interrogation signal may comprise a continuous waveform signal of a particular RF frequency.

[0103] In step **1204**, the RFID tag reader may determine if a responsive signal is received. The responsive signal may be a backscattered signal. The responsive signal may be a backscattered version of the interrogation signal transmitted in step **1202**. The responsive signal may be modulated.

[0104] If the RFID tag reader determines that a responsive signal is not received in step **1204**, step **1206** may be performed. In step **1206**, as a responsive signal was expected to be received in step **1204**, the RFID tag reader may report an error. The error may be reported to a user of the RFID tag reader. Alternatively or in addition thereto, the error may be reported to a remote user and/or a remote computing device communicatively coupled to the RFID tag reader. The error may be reported on a user interface associated with the RFID tag reader. The error may be reported using any combination of visual, audible, and/or tactile indications. After reporting the error in step **1206**, step **1202** may be repeated (e.g., if a user of the RFID tag reader determines to scan another RFID tag or the same RFID tag again).

[0105] If the RFID tag reader determines that a responsive signal is received in step **1204**, step **1208** may be performed. In step **1208**, the RFID tag reader may recover and store any data provided in the received signal. The RFID tag reader may demodulate the received signal to detect the data provided via the received signal. The data recovered may be any type of data and may include, for example, a bit sequence. The recovered bit sequence may be stored in a memory.

[0106] In step **1210**, a determination may be made (e.g., by a user of the RFID tag reader or by an automatic process controlling the RFID tag reader) as to whether the RFID tag reader is to send another interrogation signal. For example, the method of FIG. 12 may be implemented for the RFID tag reader to receive two data signals from an RFID tag. Accordingly, after transmitting a first interrogation signal and receiving first data in response thereto, the RFID tag reader may perform step **1202** to transmit a second interrogation signal in order to receive second data in response thereto. This operation may be applicable to any number of data sequences expected to be received from the RFID tag (e.g., a single data sequence or a data sequence comprising N different data signals).

[0107] If it is determined not to send another interrogation signal, step **1212** may be performed. In step **1212**, the RFID tag reader may attempt to match the stored data received from the RFID tag with a predetermined command or instruction. For example, in response to a first interrogation signal, the RFID tag reader may receive first data from the RFID tag and in response to a second

interrogation signal, the RFID tag reader may receive second data from the RFID tag. The RFID tag reader and the RFID tag may operate so as to both expect to communicate using first data and second data (as opposed to just the first data or additional data). The first data may be a binary sequence such as “010” and the second data may be a binary sequence “101.” The first data and the second data may be recovered by the RFID tag reader demodulating and/or processing a first data signal from the RFID tag and a second data signal from the RFID tag, respectively. The RFID tag may send different data signals based on a change in its orientation. The RFID tag reader may combine the data sequences to form a combined binary sequence “010101.” This data sequence may be compared to a set of stored data sequences, with each stored data sequence corresponding to a specific ID, command, instruction, request, etc. from the RFID tag. As an example, the combined binary sequence “010101” may be determined to be a valid command for authorizing a financial transaction. If the data from the RFID tag is determined to match a valid command or instruction in step **1212**, step **1214** may be performed.

[0108] In step **1214**, the RFID tag reader may forward the determined command from step **1212** to any other computing device and/or may further cause the command from the RFID tag to be implemented. For example, the RFID tag reader may forward the command to authorize a financial transaction to a POS terminal, a remote financial computing device, or any other device to which it may be communicatively coupled. The RFID tag reader may also provide one or more indications that the data from the RFID tag was successfully interpreted as a valid command. From step **1214**, step **1202** may be repeated (e.g., if it is determined that the RFID tag reader is to scan another RFID tag).

[0109] If the data from the RFID tag is not determined to match a valid command or instruction in step **1212**, step **1206** may be performed. In step **1206**, an error may be reported by the RFID tag reader to a user that a valid instruction or command was not determined. After step **1206**, step **1202** may be performed (e.g., if another attempt at scanning the tag, or scanning another RFID tag, is desired).

[0110] The method described in FIG. **12** may be used to allow the RFID tag to send data using any number of discrete data components, such that an RFID tag reader that receives the discrete data components may combine them to determine any information that the RFID intended to provide. This capability further increases the security of the RFID tag and any information stored thereon, as a bad actor would need to know a sequence of different RFID tag orientations to obtain sensitive data from the RFID tag.

[0111] Although examples are described above, features and/or steps of those examples may be combined, divided, omitted, rearranged, revised, and/or augmented in any desired manner. Various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this description, though not expressly stated herein, and are intended to be within the spirit and scope of the disclosure. Accordingly, the foregoing description is by way of example only, and is not limiting.

Claims

1. A method comprising: receiving, by a passive radio-frequency identification (RFID) tag, a radio-frequency (RF) input signal; determining, based on receiving the RF input signal, an orientation of the RFID tag; selecting, based on the determined orientation of the RFID tag, a data sequence; and transmitting an RF output signal based on the selected data sequence.
2. The method of claim 1, further comprising generating the RF output signal by modulating the RF input signal based on the selected data sequence.
3. The method of claim 1, wherein selecting the data sequence further comprises selecting the data sequence from a plurality of data sequences, each data sequence corresponding to a different orientation of the RFID tag.

4. The method of claim 1, wherein determining the orientation of the RFID tag further comprises determining the orientation of the RFID tag based on a state of a tilt switch.
5. The method of claim 1, wherein determining the orientation of the RFID tag further comprises determining the orientation of the RFID tag based on a plurality of tilt switches.
6. The method of claim 1, further comprising: receiving a second RF input signal; determining, based on receiving the second RF input signal, a second orientation of the RFID tag; selecting, based on the determined second orientation of the RFID tag, a second data sequence; and transmitting a second RF output signal based on the selected second data sequence.
7. The method of claim 6, wherein the RF output signal indicates that a first financial transaction is to be authorized and the second RF output signal indicates that a second financial transaction is unauthorized.
8. The method of claim 6, further comprising: receiving a third RF input signal; determining, based on receiving the third RF input signal, that the RFID tag is positioned in the second orientation; detecting, based on receiving the third RF input signal, a temperature associated with the RFID tag; selecting, based on the detected temperature associated with the RFID tag and based on determining that the RFID tag is positioned in the second orientation, a third data sequence; and transmitting a third RF output signal based on the selected third data sequence.
9. A radio-frequency identification (RFID) tag comprising: an antenna configured to receive a radio-frequency (RF) input signal; a power circuit configured to generate power based on the received RF input signal; one or more orientation sensors configured to detect at least a first orientation of the RFID tag and a second orientation of the RFID tag; and a controller configured to: receive the power generated by the power circuit; cause output, via the antenna, of a first RF output signal based on the first orientation of the RFID tag; and cause output, via the antenna, of a second RF output signal based on the second orientation of the RFID tag, wherein the second RF output signal is different from the first RF output signal.
10. The RFID tag of claim 9, further comprising a memory communicatively coupled to the controller and storing first data corresponding to the first orientation of the RFID tag and second data corresponding to the second orientation of the RFID tag, wherein the controller is configured to: cause the output of the first RF output signal by causing, based on detection of the first orientation and based on the first data, output of the first RF output signal; and cause the output of the second RF output signal by causing, based on detection of the second orientation and based on the second data, output of the second RF output signal.
11. The RFID tag of claim 9, further comprising a temperature detection switch communicatively coupled to the controller and configured to detect a temperature associated with the RFID tag, wherein the controller is configured to: cause the output of the first RF output signal by causing, based on detection of the first orientation and based on the detected temperature associated with the RFID tag, output of the first RF output signal; and cause the output of the second RF output signal by causing, based on detection of the second orientation and based on the detected temperature associated with the RFID tag, output of the second RF output signal.
12. The RFID tag of claim 9, wherein the RFID tag comprises a passive RFID tag.
13. The RFID tag of claim 9, wherein the one or more orientation sensors comprise a tilt switch.
14. The RFID tag of claim 9, wherein the first RF output signal and the second RF output signal each indicate one or more of: a unique identification (ID) code; a command; an instruction; or a request.
15. The RFID tag of claim 9, wherein the one or more orientation sensors are configured to detect an orientation comprising rotations about three axes.
16. A method comprising: receiving, by a radio-frequency identification (RFID) tag comprising an orientation sensor, a first radio-frequency (RF) input signal; transmitting, by the RFID tag and based a first orientation of the RFID tag, a first RF output signal; receiving, by the RFID tag, a second radio-frequency RF input signal; and transmitting, by the RFID tag and based on a second

orientation of the RFID tag, a second RF output signal, wherein the second RF output signal is different from the first RF output signal.

17. The method of claim 16, further comprising: determining, based on a first state of one or more orientation sensors, the first orientation of the RFID tag; and determining, based on a second state of the one or more orientation sensors, the second orientation of the RFID tag.

18. The method of claim 16, wherein: transmitting the first RF output signal further comprises transmitting the first RF output signal based on a detected temperature associated with the RFID tag; and transmitting the second RF output signal further comprises transmitting the second RF output signal based on the detected temperature associated with the RFID tag.

19. The method of claim 16, wherein: transmitting the first RF output signal further comprises modulating the first RF input signal based on first data corresponding to the first orientation; and transmitting the second RF output signal further comprises modulating the second RF input signal based on second data corresponding to the second orientation.

20. The method of claim 16, wherein the RFID tag is a passive RFID tag.
